

LESSON 4-5 • SESSION 2 • PRACTICE • FORM A

Solve for the Whole

Learning target – I can write and solve an equation to find the whole from a part and a percent. • 6.RP.A.3c

REMEMBER

- Let w be the whole. Write **percent $\times w = \text{part}$** , with the percent as a decimal: $60\% = 0.6$.
- Divide** to undo the multiplication: $w = \text{part} \div \text{decimal}$.
- Check: percent $\times w$ gives the part, and w is greater than the part.

WORKED EXAMPLE42 points is 35% of the team's points. Find the total w .**Step 1** $0.35 \times w = 42$ **Step 2** $w = 42 \div 0.35$ **Step 3** $w = 120$. Check: $0.35 \times 120 = 42$.**Answer 120 points****WORDS** **variable** (*variable*) • **equation** (*ecuación*) • **inverse operation** (*operación inversa*)**PRACTICE***show your work • label units***1.** Solve each equation for w .

a. $0.4 \times w = 32$ ____

b. $0.75 \times w = 45$ ____

c. $0.25 \times w = 19$ ____

d. $0.5 \times w = 34$ ____

e. $0.9 \times w = 63$ ____

f. $0.2 \times w = 13$ ____

2. Car Value. A family's car is worth \$18,900 today. That is 70% of the price they paid for it.**Hint** *Today's value is the part; the price paid is the whole.***a.** Write an equation using w for the price paid. _____**b.** Solve: $w =$ _____*Show your work.***3. Concert Ticket.** A balcony ticket costs \$84. That is 40% of the front-row price. Find the front-row price.*Show your work.***Price** _____**4. Practice Minutes.** Kai practiced 96 minutes. That is 80% of his weekly goal.**a.** Goal _____**b.** Minutes still needed _____*Show your work.***5. Fundraiser.** A team has raised \$312. That is 65% of its goal. Write and solve an equation to find the goal.**Hint** *Divide the part by 0.65.**Show your work.***Goal** _____

- 6. Equation Claim.** A shirt costs \$27. That is 75% of its first price. Dev wrote $w = 0.75 \times 27$. Is his equation right? Write and solve a correct one.

Hint Which number is the part?

Show your work.

First price _____

- 7. Monthly Income.** A family pays \$1,050 in rent each month. That is 35% of their monthly income. Find the income. How much is left after rent?

Hint Solve for the income first, then subtract the rent.

Show your work.

Income, left _____

- 8. Two Discounts.** A jacket sold for \$63, which is 70% of its first price. Boots sold for \$63, which is 45% of their first price.

- a. Write and solve an equation for each first price. _____
- b. Which item was discounted more, and by how many dollars? _____
- c. Why did the same sale price come from two different first prices?

FIND THE ERROR

explain the mistake, then fix it

- 9. Sofia's Work.** 25% of a whole is \$52. Sofia wrote: " $w = 0.25 \times 52 = \$13$." Explain her mistake. Then write and solve the correct equation.

TEST PRACTICE

choose the best answer

- 10.** Which equation finds the whole when 30 is 20% of it?
A $0.2 \times 30 = w$ **B** $0.2 \times w = 30$ **C** $30 \times w = 0.2$ **D** $20 \times w = 30$
- 11.** Solving $45 = 0.75w$ gives $w =$ –
A 33.75 **B** 45.75 **C** 60 **D** 75

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

\$480 I	\$20.25 Q	\$210 I	\$33.60 Z	24 V	\$36 D	27,000 D	\$3,000 E	\$13 X
#2b		#3	#4b	#5	#6	#7		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____